



40.1

Binary Tree (1)

40.1.1 Binary Tree: NIELIT 2021 Dec Scientist B - Section B: 120



Let B_n denote the number of full binary trees with n vertices. Then a recurrence relation for B_n is :

A. $B_n = B_{n-1} + O(1)$
 C. $B_n = B_{n-1} + O(n)$

B. $B_n = 2B_{n-1} + O(1)$
 D. $B_n = 2B_{n-1} + O(n)$

nielit2021dec-scientistb recurrence-relation binary-tree data-structures algorithms

40.2

Complexity Classes (2)

40.2.1 Complexity Classes: NIELIT 2021 Dec Scientist A - Section B: 74



Assume that P and NP are different i.e. $P \neq NP$ then for the expression $NP\text{-Complete} \cap P = ?$ Which among the following is correct ?

A. NP-Hard

B. $\emptyset$

C. P

D. NP-Complete

nielit2021dec-scientista theory-of-computation complexity-classes

Answer key

40.2.2 Complexity Classes: NIELIT 2021 Dec Scientist B - Section B: 46



_____ is the class of decision problems that can be solved by non-deterministic polynomial algorithms.

A. NP

B. P

C. Hard

D. Complete

nielit2021dec-scientistb theory-of-computation p-np-npc-nph complexity-classes

Answer key

40.3

Complexity Theory (1)

40.3.1 Complexity Theory: NIELIT 2021 Dec Scientist A - Section B: 109



Given a graph with n vertices, deciding if there exists a clique of size ≥ 195 is :

A. Solvable in polynomial time

B. NP

C. NP-Complete

D. None of the above

nielit2021dec-scientista graph-theory complexity-theory p-np-npc-nph algorithm-design

Answer key

40.4

Context Free Grammar (3)

40.4.1 Context Free Grammar: NIELIT 2016 MAR Scientist B - Section C: 29



The CFG $S \rightarrow aS \mid bS \mid a \mid b$ is equivalent to

- A. $(a + b)$ B. $(a + b)(a + b)^*$ C. $(a + b)(a + b)$ D. all of these

nielit2016mar-scientistb theory-of-computation context-free-grammar

Answer key

40.4.2 Context Free Grammar: NIELIT 2017 DEC Scientist B - Section B: 7



Let G be a grammar in CFG and let $W_1, W_2 \in L(G)$ such that $|W_1| = |W_2|$ then which of the following statements is true?

- A. Any derivation of W_1 has exactly the same number of steps as any derivation of W_2 .
B. Different derivation have different length.
C. Some derivation of W_1 may be shorter than the derivation of W_2
D. None of the options

nielit2017dec-scientistb theory-of-computation context-free-grammar

Answer key

40.4.3 Context Free Grammar: NIELIT 2021 Dec Scientist A - Section B: 98



Which of the following problems is undecidable ?

- A. Membership problem for CFGs. B. Ambiguity problem for CFGs.
C. Finiteness problem for FSAs. D. Equivalence problem for FSAs.

nielit2021dec-scientista theory-of-computation decidability context-free-grammar finite-automata

Answer key

40.5 Context Free Language (5)

40.5.1 Context Free Language: NIELIT 2016 DEC Scientist B (CS) - Section B: 3



If L_1 is CFL and L_2 is regular language which of the following is false?

- A. $L_1 - L_2$ is not Context free B. L_1 intersection L_2 is Context free
C. $\sim L_1$ is Context free D. Both (A) and (C)

nielit2016dec-scientistb-cs theory-of-computation context-free-language

Answer key

40.5.2 Context Free Language: NIELIT 2016 MAR Scientist B - Section C: 26



If L_1 and L_2 are context free language and R a regular set, then which one of the languages below is not necessarily a context free language?

- A. $L_1 L_2$ B. $L_1 \cap L_2$ C. $L_1 \cap R$ D. $L_1 \cup L_2$

nielit2016mar-scientistb theory-of-computation context-free-language

Answer key

40.5.3 Context Free Language: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 31



Which of the following definitions generates the same languages as L , where $L = \{x^n y^n, n \geq 1\}$

- i. $E \rightarrow xEy \mid xy$
- ii. $xy \mid x^+ xyy^+$
- iii. $x^+ y^+$

A. (i) B. (i) and (ii) only C. (ii) and (iii) only D. (ii) only

nielit2017oct-assistanta-cs theory-of-computation context-free-language

Answer key

40.5.4 Context Free Language: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 9



Which of the following definitions generates the same languages as L , where $L = \{x^n y^n, n \geq 1\}$

- i. $E \rightarrow xEy \mid xy$
- ii. $xy \mid x^+ xyy^+$
- iii. $x^+ y^+$

A. (i) Only B. (i) and (ii) only C. (ii) and (iii) only D. (ii) only

nielit2017oct-assistanta-it theory-of-computation context-free-language

Answer key

40.5.5 Context Free Language: NIELIT 2018-33



The language $\{W^a X^b Y^{a+b} \mid a, b, > 1\}$ is

- A. Regular
- B. Context-free but not regular
- C. Context sensitive but not context free
- D. Type= 0 but not context sensitive

nielit-2018 theory-of-computation context-free-language

Answer key

40.6

Context Sensitive (1)

40.6.1 Context Sensitive: NIELIT 2021 Dec Scientist B - Section B: 80



The following grammar is an example of _____ .

$A \rightarrow a A B C$

$CB \rightarrow B c$

$A \rightarrow a b c$

$b B \rightarrow b b$

$B \rightarrow c b a$

A. Type 0 Grammar

B. Type 1 Grammar

C. Type 2 Grammar

D. Type 3 Grammar

nielit2021dec-scientistb grammar theory-of-computation context-sensitive

40.7

Cyk Algorithm (1)

40.7.1 Cyk Algorithm: NIELIT 2017 DEC Scientist B - Section B: 50



Let n is the length of string to test for membership, then the number of table entry in CYK algorithm is

A. $n(n+1)$

B. $n^2 + 1$

C. $n^2 - 1$

D. $n(n+1)/2$

nielit2017dec-scientistb theory-of-computation cyk-algorithm

Answer key

40.8

Decidability (2)

40.8.1 Decidability: NIELIT 2017 DEC Scientist B - Section B: 11



Which of the following are undecidable?

P1: The language generated by some CFG contains any words of length less than some given number n .

P2: Let $L1$ be CFL and $L2$ be regular, to determine whether $L1$ and $L2$ have common elements

P3: Any given CFG is ambiguous or not.

P4: For any given CFG G , to determine whether epsilon belongs to $L(G)$

A. *P2* only

B. *P1* and *P2* only

C. *P2* and *P3* only

D. *P3* only

nielit2017dec-scientistb theory-of-computation decidability

Answer key

40.8.2 Decidability: NIELIT 2022 April Scientist B | Section B | Question: 108



Consider the following problem X .

Given a Turing machine M over the input alphabet Σ , any state q of M and a word $w \in \Sigma^*$, does the computation of M on w visit the state of q ?

Which of the following statements about X is correct?

- A. X is decidable
 B. X is undecidable but partially decidable
 C. X is undecidable and not even partially decidable
 D. X is not a decision problem

nielit2022apr-scientistb theory-of-computation turing-machine decidability

40.9

Dpda (2)

40.9.1 Dpda: NIELIT 2017 DEC Scientific Assistant A - Section B: 6



Which of the following statements is true ?

- A. Melay and Moore machines are language acceptors.
 B. Finite State automata is language translator.
 C. NPDA is more powerful than DPDA.
 D. Melay machine is more powerful than Moore machine.

nielit2017dec-assistanta theory-of-computation dpda npda

Answer key

40.9.2 Dpda: NIELIT 2017 DEC Scientist B - Section B: 39



Which of the following is **true**?

- A. Mealy and Moore machine are language acceptors.
 B. Finite State automata is language translator.
 C. NPDA is more powerful than DPDA.
 D. Melay machine is more powerful than Moore machine.

nielit2017dec-scientistb theory-of-computation finite-automata npda dpda

Answer key

40.10

Expression Evaluation (2)

40.10.1 Expression Evaluation: NIELIT 2021 Dec Scientist A - Section B: 87



The postfix equivalent of the infix expression $(a + b)^*(c^*d - e)^*f/g$ is :

- A. $ab + cd^*e - fg^*/^*$
 B. $ab + cd^*e - fg/**$
 C. $ab + cd^*e - fg/**$
 D. $abcd + e^*fg/**$

nielit2021dec-scientista expression-evaluation stack data-structures compiler-design

Answer key

40.10.2 Expression Evaluation: NIELIT 2021 Dec Scientist B - Section B: 60



$ABC*+$ is the postfix form of :

- A. $A * B + C$
 B. $A * + BC$
 C. $A + B * C$
 D. none of these

Answer key 

40.11

Finite Automata (10)

40.11.1 Finite Automata: NIELIT 2016 DEC Scientist B (CS) - Section B: 1



Palindromes can't be recognized by any Finite State Automata because:

- A. FSA cannot remember arbitrarily large amount of information.
- B. FSA cannot deterministically fix the midpoint.
- C. Even if the mid-Point is known an FSA cannot find whether the second half of the string matches the first half.
- D. All of the above.

nielit2016dec-scientistb-cs theory-of-computation finite-automata

Answer key 

40.11.2 Finite Automata: NIELIT 2016 DEC Scientist B (CS) - Section B: 24



Given two DFA's $M1$ and $M2$. They are equivalent if

- A. $M1$ and $M2$ has the same number of states
- B. $M1$ and $M2$ accepts the same language i.e $L(M1) = L(M2)$
- C. $M1$ and $M2$ has the same number of final states
- D. None of the above

nielit2016dec-scientistb-cs theory-of-computation finite-automata

Answer key 

40.11.3 Finite Automata: NIELIT 2016 DEC Scientist B (CS) - Section B: 46



$(00 + 01 + 10)(0 + 1)^*$ represents

- A. Strings not starting with 11
- B. Strings of odd length
- C. Strings starting with 00
- D. Strings of even length

nielit2016dec-scientistb-cs theory-of-computation finite-automata

Answer key 

40.11.4 Finite Automata: NIELIT 2016 MAR Scientist C - Section C: 32



Two finite state machines are said to be equivalent if they

- A. have same number of states
- B. have same number of edges
- C. have same number of states and edges
- D. recognize same set of tokens

nielit2016mar-scientistc theory-of-computation finite-automata

Answer key 

40.11.5 Finite Automata: NIELIT 2017 DEC Scientific Assistant A - Section B: 3

The automaton which allows transformation to a new state without consuming any symbols :



- A. *NFA* B. *DFA* C. *NFA* – 1 D. All of the options

nielit2017dec-assistanta theory-of-computation finite-automata

Answer key

40.11.6 Finite Automata: NIELIT 2017 DEC Scientific Assistant A - Section B: 4

Complement of a *DFA* can be obtained by :



- A. making starting state as final state. B. make final as a starting state.
C. making final states non-final and D. None of the options
non-final as final.

nielit2017dec-assistanta theory-of-computation finite-automata

Answer key

40.11.7 Finite Automata: NIELIT 2017 DEC Scientific Assistant A - Section B: 40

What is the relation between *DFA* and *NFA* on the basis of computational power ?



- A. *DFA* > *NFA* B. *NFA* > *DFA* C. Equal D. Can't be said

nielit2017dec-assistanta theory-of-computation finite-automata

Answer key

40.11.8 Finite Automata: NIELIT 2017 DEC Scientific Assistant A - Section B: 5

Concatenation Operation refers to which of the following set operations :



- A. Union B. Dot C. Kleene D. None of the options

nielit2017dec-assistanta theory-of-computation finite-automata

Answer key

40.11.9 Finite Automata: NIELIT 2017 DEC Scientific Assistant A - Section B: 60

Finite automata requires minimum _____ number of stacks.



- A. 1 B. 0 C. 2 D. None of the options

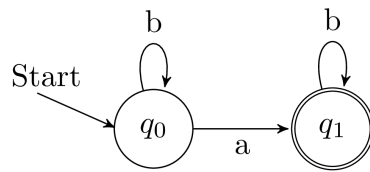
nielit2017dec-assistanta theory-of-computation finite-automata

Answer key

40.11.10 Finite Automata: NIELIT 2018-40



Consider the following Finite State Automaton. The language accepted by the automaton is given by the regular expression.



- A. ab^*b^* B. a^*b^* C. b^*b D. b^*ab^*

nielit-2018 theory-of-computation finite-automata

Answer key

40.12

First Order Logic (1)

40.12.1 First Order Logic: NIELIT 2021 Dec Scientist B - Section B: 49



Let $P(x)$ be “ x is perfect”, $F(x)$ be “ x is your friend” and the domain be all people. The statement, “At least one of your friends is perfect” is :

- A. $\forall x (F(x) \rightarrow P(x))$
B. $\forall x (F(x) \wedge P(x))$
C. $\exists x (F(x) \wedge P(x))$
D. $\exists x (F(x) \rightarrow P(x))$

nielit2021dec-scientistb propositional-logic first-order-logic logical-reasoning

Answer key

40.13

Grammar (2)

40.13.1 Grammar: NIELIT 2021 Dec Scientist A - Section B: 67



_____ is the most general phase structured grammar.

- A. Regular B. Context free
C. Context sensitive D. All of the above

nielit2021dec-scientista grammar theory-of-computation

Answer key

40.13.2 Grammar: NIELIT 2021 Dec Scientist B - Section B: 112



$(a+b)^2$ corresponds to the language :

- A. $\{a+b, a+b\}$ B. $\{aa, ab, ba, bb\}$
C. $\{abab, baba\}$ D. $\{a+b, (a+b)^2\}$

nielit2021dec-scientistb theory-of-computation regular-expression grammar

Answer key 

40.14

Identify Class Language (7)

40.14.1 Identify Class Language: NIELIT 2016 MAR Scientist B - Section C: 25

Regarding power of recognition of language, which of the following statements is false?



- A. Non deterministic finite-state automata are equivalent to deterministic finite-state automata.
- B. Non-deterministic push-down automata are equivalent to deterministic push-down automata.
- C. Non-deterministic Turing Machines are equivalent to deterministic push-down automata.
- D. Multi-tape Turing Machines are equivalent to Single-tape Turing Machines.

nielit2016mar-scientistb theory-of-computation identify-class-language

Answer key 

40.14.2 Identify Class Language: NIELIT 2016 MAR Scientist C - Section C: 14

If every string of a language can be determined, whether it is legal or illegal in finite time, the language is called



- A. decidable B. undecidable C. interpretive D. non-deterministic

nielit2016mar-scientistc theory-of-computation identify-class-language

Answer key 

40.14.3 Identify Class Language: NIELIT 2016 MAR Scientist C - Section C: 15

The defining language for developing a formalism in which language definitions can be stated, is called



- A. syntactic meta language B. decidable language
C. intermediate language D. high level language

nielit2016mar-scientistc theory-of-computation identify-class-language non-gatecse

Answer key 

40.14.4 Identify Class Language: NIELIT 2017 DEC Scientific Assistant A - Section B: 31

A finite automaton accepts which type of language :



- A. Type 0 B. Type 1 C. Type 2 D. Type 3

nielit2017dec-assistanta theory-of-computation identify-class-language finite-automata

Answer key 

40.14.5 Identify Class Language: NIELIT 2017 DEC Scientist B - Section B: 56



Which of the following statement is **true**?

- A. Deterministic context free language are closed under complement.
- B. Deterministic context free language are not closed under Union.
- C. Deterministic context free language are closed under intersection with regular set.
- D. All of the options

nielit2017dec-scientistb theory-of-computation identify-class-language context-free-language

Answer key

40.14.6 Identify Class Language: NIELIT 2017 DEC Scientist B - Section B: 58



Which of the following is a correct hierarchical relationships of the following where

L_1 : set of languages accepted by NFA

L_2 : set of languages accepted by DFA

L_3 : set of languages accepted by DPDA

L_4 : set of languages accepted by NPDA

L_5 : set of recursive language

L_6 : set of recursive enumerable languages?

- A. $L_1, L_2 \subset L_3 \subset L_4 \subset L_5 \subset L_6$
- B. $L_1 \subset L_2 \subset L_3 \subset L_4 \subset L_5 \subset L_6$
- C. $L_2 \subset L_1 \subset L_3 \subset L_4 \subset L_5 \subset L_6$
- D. $L_1 \subset L_2 \subset L_3 \subset L_4 \subset L_6 \subset L_5$

nielit2017dec-scientistb theory-of-computation identify-class-language recursive-and-recursively-enumerable-languages

Answer key

40.14.7 Identify Class Language: NIELIT 2022 April Scientist B | Section B | Question: 92



L_1 is a recursively enumerable language over Σ . An algorithm A effectively enumerates its words as $w_1, w_2, w_3, \dots$. Define another language L_2 over $\Sigma \cup \{\#\}$ as $\{wi\#wj : wi, wj \in L_1, i < j\}$. Here $\#$ is new symbol. Consider the following assertions.

- $S_1 : L_1$ is recursive implies L_2 is recursive
- $S_2 : L_2$ is recursive implies L_1 is recursive

Which of the following statements is true?

- A. Both S_1 and S_2 are true
- B. S_1 is true but S_2 is not necessarily true
- C. S_2 is true but S_1 is not necessarily true
- D. Neither is necessarily true

nielit2022apr-scientistb theory-of-computation recursive-and-recursively-enumerable-languages decidability identify-class-language

40.15

Injective Function (1)

40.15.1 Injective Function: NIELIT 2022 April Scientist B | Section B | Question: 82

Consider the following two statements:



- i. A hash function (these are often used for computing digital signatures) is an injective function.
- ii. encryption technique such as DES performs a permutation on the elements of its input alphabet.

Which one of the following options is valid for the above two statements?

- | | |
|--|---|
| A. Both are false | B. Statement (i) is true and the other is false |
| C. Statement (ii) is true and the other is false | D. Both are true |

nielit2022apr-scientistb hashing encryption-decryption injective-function digital-signature combinatory cryptography

40.16

Language Classes (1)

40.16.1 Language Classes: NIELIT 2021 Dec Scientist A - Section B: 99

Let $L = L_1 \cap L_2$ where L_1 and L_2 are language defined below :

$$L_1 = \{a^m b^m c a^n b^n \mid m, n > 0\}$$

$$L_2 = \{a^i b^j c^k \mid i, j, k > 0\}$$

Then L is

- | | |
|---------------------------------|--|
| A. Not Recursive | B. Regular |
| C. Context free but not regular | D. Recursively enumerable but not context free |

nielit2021dec-scientista theory-of-computation context-free-language recursively-enumerable-languages language-classes

40.17

Lexical Analysis (1)

40.17.1 Lexical Analysis: NIELIT Scientist B 2020 November: 57

Which of the following machine model can be used in a necessary and sufficient sense for lexical analysis in modern computer language?



- | | |
|--------------------------------------|--------------------|
| A. Deterministic Push down Automata | B. Finite Automata |
| C. Non-Deterministic Finite Automata | D. Turing Machine |

nielit-scb-2020 finite-automata lexical-analysis compiler-design

Answer key

40.18

Number of Dfa (1)

40.18.1 Number of Dfa: NIELIT 2017 DEC Scientific Assistant A - Section B: 50

How many DFA's exists with two states over input alphabet $\{0, 1\}$



- A. 16 B. 26 C. 32 D. 64

nielit2017dec-assistanta theory-of-computation finite-automata number-of-dfa

Answer key

40.19

P NP NPC NPH (2)

40.19.1 P NP NPC NPH: NIELIT 2016 MAR Scientist B - Section C: 19



If there is in NP-Complete language L whose complement is in NP, then complement of any language in NP is in

- A. P B. NP C. both (A) and (B) D. None of these

nielit2016mar-scientistb theory-of-computation p-np-npc-nph

Answer key

40.19.2 P NP NPC NPH: NIELIT Scientific Assistant A 2020 November: 57



Which of the following problem is not NP complete but undecidable?

- A. Partition Problem B. Halting Problem
C. Hamiltonian Circuit D. Bin Packing

nielit-sta-2020 theory-of-computation p-np-npc-nph

Answer key

40.20

Pumping Lemma (1)

40.20.1 Pumping Lemma: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 12



The logic of pumping lemma is a good example of

- A. the pigeon-hole principle B. the divide and conquer technique
C. recursion D. iteration

nielit2017oct-assistanta-cs theory-of-computation pumping-lemma

Answer key

40.21

Pushdown Automata (1)

40.21.1 Pushdown Automata: NIELIT 2017 DEC Scientist B - Section B: 57



Which machine is equally powerful in both deterministic and non-deterministic form?

- A. Push Down Automata B. Turing machine
C. Linear Bounded Automata D. None of the options

nielit2017dec-scientistb pushdown-automata turing-machine

Answer key 

40.22

Quantifiers (1)

40.22.1 Quantifiers: NIELIT 2021 Dec Scientist A - Section B: 43



Let the predicates $D(x, y)$ mean “team x defeated team y ” and $P(x, y)$ mean “team x has played team y ”. The quantified formula for the statement that there is a team that has beaten every team it has played is:

- A. $\exists x \forall y (P(x, y) \rightarrow D(x, y))$
- B. $\forall x \exists y (P(x, y) \rightarrow D(x, y))$
- C. $\forall y \exists x (P(x, y) \rightarrow D(x, y))$
- D. $\exists x \forall y (D(x, y) \rightarrow P(x, y))$

nielit2021dec-scientista first-order-logic logical-reasoning quantifiers

Answer key 

40.23

Recursion (1)

40.23.1 Recursion: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 32



Choose the correct statements.

- A. A total recursive function is also a partial recursive function
- B. A partial recursive function is also a total recursive function
- C. A partial recursive function is also a primitive recursive function
- D. None of the above

nielit2017oct-assistanta-cs theory-of-computation recursion

Answer key 

40.24

Recursive and Recursively Enumerable Languages (9)

40.24.1 Recursive and Recursively Enumerable Languages: NIELIT 2017 DEC Scientist B - Section B: 19



Recursive enumerable languages are not closed under _____.

- A. Set difference
- B. Complement
- C. Both (A) and (B)
- D. None of the options

nielit2017dec-scientistb theory-of-computation easy recursive-and-recursively-enumerable-languages

Answer key 

40.24.2 Recursive and Recursively Enumerable Languages: NIELIT 2017 DEC Scientist B - Section B: 27



If any string of a language L can be effectively enumerated by an enumerator in a

lexicographic order then language L is _____.

- A. Regular
- B. Context free but not necessarily regular
- C. Recursive but not necessarily context free
- D. Recursively enumerable but not necessarily recursive

nielit2017dec-scientistb theory-of-computation recursive-and-recursively-enumerable-languages

Answer key 

40.24.3 Recursive and Recursively Enumerable Languages: NIELIT 2017 DEC Scientist B - Section B: 32



The collection of Turing recognizable languages are closed under:

- i. Union
- ii. Intersection
- iii. Complement
- iv. Concatenation
- v. Star Closure

- A. (i) Only
- B. Both (i),(iv)
- C. (i),(ii),(iv)and(v)
- D. All of the options

nielit2017dec-scientistb theory-of-computation easy recursive-and-recursively-enumerable-languages

Answer key 

40.24.4 Recursive and Recursively Enumerable Languages: NIELIT 2017 July Scientist B (CS) - Section B: 59



Let L be a language and L' be its complement. Which one of the following is NOT a viable possibility?

- A. Neither L nor L' is RE.
- B. One of the L and L' is RE but not recursive;the other is not RE.
- C. Both L and L' are RE but not recursive.
- D. Both L and L' are recursive.

nielit2017july-scientistb-cs theory-of-computation recursive-and-recursively-enumerable-languages

Answer key 

40.24.5 Recursive and Recursively Enumerable Languages: NIELIT 2017 July Scientist B (CS) - Section B: 60



Let $L1$ be a recursive language, and let $L2$ be a recursively enumerable but not recursive language. Which one of the following is TRUE?

- A. $L1'$ is recursive and $L2'$ is recursively enumerable.
- B. $L1'$ is recursive and $L2'$ is not recursively enumerable.
- C. $L1'$ and $L2'$ is recursively enumerable.

D. $L1'$ is recursively enumerable and $L2'$ is recursive.

nielit2017july-scientistb-cs theory-of-computation recursive-and-recursively-enumerable-languages

Answer key 

40.24.6 Recursive and Recursively Enumerable Languages: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 9



A language L for which there exists a $TM 'T'$, that accepts every word in L and either rejects or loops for every word that is not in L , is said to be

- A. Recursive
- B. Recursively enumerable
- C. NP-HARD
- D. None of the above

nielit2017oct-assistanta-cs theory-of-computation recursive-and-recursively-enumerable-languages

Answer key 

40.24.7 Recursive and Recursively Enumerable Languages: NIELIT 2018-81



Identify the true statement from the given statements:

1. A recursive formal language is a recursive subset in the set of all possible words over the alphabet of the language.
2. The complement of a recursive languages is recursive
3. The complement of a context-free language is context-free

- A. Only 1
- B. 1 and 2
- C. 1, 2 and 3
- D. 2 and 3

nielit-2018 theory-of-computation recursive-and-recursively-enumerable-languages

Answer key 

40.24.8 Recursive and Recursively Enumerable Languages: NIELIT 2021 Dec Scientist B - Section B: 52



A language L is recognizable by a turing machine M if and only if L is a _____ language.

- A. Type 0
- B. Type 1
- C. Type 2
- D. Type 3

nielit2021dec-scientistb theory-of-computation turing-machine recursive-and-recursively-enumerable-languages

Answer key 

40.24.9 Recursive and Recursively Enumerable Languages: NIELIT 2022 April Scientist B | Section B | Question: 43



Consider the following types of languages:

- $L1$: Regular,
- $L2$: Context-free,
- $L3$: Recursive,

L_4 : Recursively enumerable.

Which of the following is/are TRUE ?

- I. $L_3' \cup L_4$ is recursively enumerable
- II. $L_2 \cup L_3$ is recursive
- III. $L_1^* \cup L_2$ is context-free
- IV. $L_1 \cup L_2'$ is context-free

- A. I only
- B. I and III only
- C. I and IV only
- D. I, II and III only

nielit2022apr-scientistb theory-of-computation recursive-and-recursively-enumerable-languages

Answer key

40.25

Regular Expression (13)

40.25.1 Regular Expression: NIELIT 2016 MAR Scientist B - Section C: 24



Which of the following regular expressions denotes a language comprising all possible strings over the alphabet $\{a, b\}$?

- A. a^*b^*
- B. $(a | b)^*$
- C. $(ab)^+$
- D. $(a | b^*)$

nielit2016mar-scientistb theory-of-computation regular-expression

Answer key

40.25.2 Regular Expression: NIELIT 2016 MAR Scientist C - Section C: 17



Regular expression $(a | b)(a | b)$ denotes the set

- A. $\{a, b, ab, aa\}$
- B. $\{a, b, ba, bb\}$
- C. $\{a, b\}$
- D. $\{aa, ab, ba, bb\}$

nielit2016mar-scientistc theory-of-computation regular-expression

Answer key

40.25.3 Regular Expression: NIELIT 2017 DEC Scientific Assistant A - Section B: 19



Let P, Q, R be a regular expression over Σ . If P does not contain null string, then $R = Q + RP$ has a unique solution _____.

- A. Q^*P
- B. QP^*
- C. Q^*P^*
- D. $(P^*Q)^*$

nielit2017dec-assistanta theory-of-computation regular-expression

Answer key

40.25.4 Regular Expression: NIELIT 2017 DEC Scientific Assistant A - Section B: 35



$(0 + \epsilon)(1 + \epsilon)$ represents :

- A. $\{0, 1, 01, \epsilon\}$
- B. $\{0, 1, \epsilon\}$

C. $\{0, 1, 01, 11, 00, 10, \varepsilon\}$

D. $\{0, 1, \}$

nielit2017dec-assistanta theory-of-computation regular-expression

Answer key

40.25.5 Regular Expression: NIELIT 2017 DEC Scientific Assistant A - Section B: 54

Complement of $(a + b)^*$ will be :



A. Φ

B. Null

C. a

D. b

nielit2017dec-assistanta theory-of-computation regular-expression

Answer key

40.25.6 Regular Expression: NIELIT 2017 DEC Scientist B - Section B: 10

A regular expression is $(a + b^*c)$ is equivalent to



A. set of strings with either a or one or more occurrence of b followed by c .

B. $(b^*c + a)$

C. set of strings with either a or zero or more occurrence of b followed by c .

D. Both (B) and (C)

nielit2017dec-scientistb theory-of-computation regular-expression

Answer key

40.25.7 Regular Expression: NIELIT 2017 DEC Scientist B - Section B: 2

According to the given language, which among the following expressions does it correspond to ?



Language $L = \{x \in \{0, 1\} \mid x \text{ is of length 4 or less}\}$.

A. $(0 + 1 + 0 + 1 + 0 + 1 + 0 + 1)^4$

B. $(0 + 1)^4$

C. $(01)^4$

D. $(0 + 1 + \varepsilon)^4$

nielit2017dec-scientistb theory-of-computation regular-expression

Answer key

40.25.8 Regular Expression: NIELIT 2017 DEC Scientist B - Section B: 22

What is the meaning of regular expression $\Sigma^*001\Sigma^*$?



A. Any string containing '1' as substring

B. Any string containing '01' as substring

C. Any string containing '011' as substring

D. All string containing '001' as substring

nielit2017dec-scientistb theory-of-computation regular-expression

Answer key

Which of the following regular expression is equal to $(r_1 + r_2)^*$?

- A. $r_1^* r_2^*$
B. $(r_1 r_2)^*$
C. $r_1^* r_2^* + r_1 r_2$
D. $(r_1^* r_2^*)^*$

nielit2017dec-scientistb theory-of-computation regular-expression

Answer key

Which of the following is equivalent regular expressions?

- i. $((01)^*(10)^*)^*$
- ii. $(10 + 01)^*$
- iii. $(01)^* + (11)^*$
- iv. $(0^* + (11)^* + 0^*)^*$

- A. (i) and (ii) B. (ii) and (iii) C. (iii) and (iv) D. (iv) and (i)

nielit2017dec-scientistb theory-of-computation regular-expression

Answer key

The string 1101 does not belong to the set represented by

- A. $(00 + (11)^*0)$
 B. $1(0 + 1)^*101$
 C. $(10)^*(01)^*(00 + 11)^*$
 D. $110^*(0 + 1)$

nielit2017dec-scientistb theory-of-computation regular-expression

Answer key

Which of the following languages over the alphabet $[0, 1]$ is described by the given regular expression: $(0 + 1)^*1(0 + 1)^*1$?

- A. The set of all strings containing the substring 11
B. The set of all strings containing at most two 1's
C. The set of all strings containing at least two 1's
D. The set of all strings that begins and ends with only 0

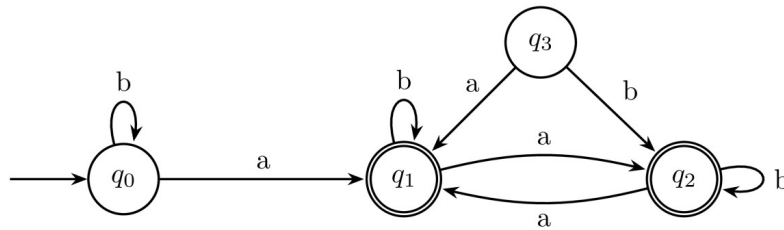
nielit-2018 theory-of-computation regular-expression

Answer key

40.25.13 Regular Expression: NIELIT 2021 Dec Scientist A - Section B: 103



Consider the finite automata given below



The language b accepted by this automata is a given by the regular expression :

- A. $b^*ab^*ab^*ab^*$ B. $(a+b)^*$ C. $b^*a(a+b)^*$ D. $b^*ab^*ab^*$

nielit2021dec-scientista finite-automata regular-expression theory-of-computation

Answer key

40.26

Regular Language (5)

40.26.1 Regular Language: NIELIT 2016 MAR Scientist B - Section C: 27



If L be a language recognizable by a finite automaton, then language from $\{L\} = \{w \text{ such that } w \text{ is a prefix of } v \text{ where } v \in L\}$, is a

- A. regular language. B. context-free language.
C. context-sensitive language. D. recursive enumeration language

nielit2016mar-scientistb theory-of-computation regular-language

Answer key

40.26.2 Regular Language: NIELIT 2016 MAR Scientist B - Section C: 28



Which of the following statements is correct?

- A. $A = \{a^n b^n \mid n = 0, 1, 2, 3, \dots\}$ is regular language
B. Set B of all strings of equal number of a 's and b 's defines a regular language
C. $L(A^*B^*) \cap B$ gives the set A
D. None of these.

nielit2016mar-scientistb theory-of-computation regular-language

Answer key

40.26.3 Regular Language: NIELIT 2017 DEC Scientific Assistant A - Section B: 15



If $L1$ and $L2$ are regular sets then intersection of these two will be :

- A. Regular B. Non Regular C. Recursive D. Non Recursive

nielit2017dec-assistanta theory-of-computation regular-language

Answer key

40.26.4 Regular Language: NIELIT 2018-83



In the Given language $L = \{ab, aa, baaa\}$, ____ number of strings are in L^*

1. baaaba
2. aabaaaa
3. baaabaaaabaa
4. baaabaaa

A. 1

B. 2

C. 3

D. 4

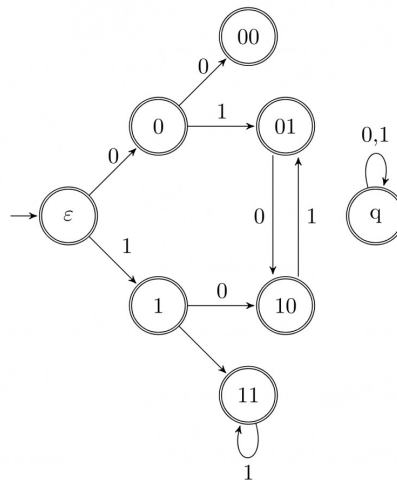
nielit-2018 theory-of-computation regular-language

Answer key

40.26.5 Regular Language: NIELIT 2022 April Scientist B | Section B | Question: 87



Consider the set of strings on $\{0, 1\}$ in which, every substring of 3 symbols has at most two zeros. For example, 001110 and 011001 are in the language, but 100010 is not. All strings of length less than 3 are also in the language. A partially completed DFA that accepts this language is shown below.



The missing arcs in the DFA are:

A.

	00	01	10	11	q
00	1	0			
01				1	
10	0				
11			0		

		00	01	10	11	q
	00		0			1
B.	01		1			
	10				0	
	11		0			
		00	01	10	11	q
	00		1			0
C.	01		1			
	10			0		
	11		0			
		00	01	10	11	q
	00		1			0
D.	01				1	
	10	0				
	11			0		

nielit2022apr-scientistb theory-of-computation finite-automata regular-language

40.27

Turing Machine (3)

40.27.1 Turing Machine: NIELIT 2016 DEC Scientist B (CS) - Section B: 6



Which of the following is wrong?

- A. Turing machine is a simple mathematical model of general purpose computer
- B. Turing machine is more powerful than finite automata
- C. Turing Machine can be simulated by a general purpose computer
- D. All of these

nielit2016dec-scientistb-cs theory-of-computation turing-machine

Answer key 

40.27.2 Turing Machine: NIELIT 2017 DEC Scientist B - Section B: 37



Which of the following statement is true?

S1: The power of a multi-tape Turing machine is greater than the power of a single tape Turing machine.

S2: Every non-deterministic Turing machine has an equivalent deterministic Turing machine.

- A. *S1*
- B. *S2*
- C. Both *S1* and *S2*
- D. None of the options

Answer key

40.27.3 Turing Machine: NIELIT 2021 Dec Scientist A - Section B: 91



If T_1 and T_2 are two Turing machines. The composite can be represented using the expression :

- A. T_1T_2 B. $T_1 \cup T_2$ C. $T_1 \times T_2$ D. None of the options

Answer key

Answer Keys

40.1.1	B	40.2.1	B	40.2.2	A	40.3.1	C	40.4.1	B
40.4.2	C	40.4.3	B	40.5.1	D	40.5.2	B	40.5.3	A
40.5.4	A	40.5.5	B	40.6.1	B	40.7.1	D	40.8.1	D
40.8.2	B	40.9.1	C	40.9.2	C	40.10.1	TBA	40.10.2	C
40.11.1	D	40.11.2	B	40.11.3	A	40.11.4	D	40.11.5	C
40.11.6	C	40.11.7	C	40.11.8	B	40.11.9	B	40.11.10	D
40.12.1	C	40.13.1	C	40.13.2	B	40.14.1	B	40.14.2	A
40.14.3	A	40.14.4	D	40.14.5	D	40.14.6	A	40.14.7	A
40.15.1	C	40.16.1	B	40.17.1	B	40.18.1	D	40.19.1	B
40.19.2	B	40.20.1	A	40.21.1	B	40.22.1	A	40.23.1	A
40.24.1	C	40.24.2	C	40.24.3	C	40.24.4	C	40.24.5	B
40.24.6	B	40.24.7	B	40.24.8	A	40.24.9	D	40.25.1	B
40.25.2	D	40.25.3	B	40.25.4	A	40.25.5	A	40.25.6	D
40.25.7	D	40.25.8	D	40.25.9	D	40.25.10	A	40.25.11	C
40.25.12	X	40.25.13	C	40.26.1	A	40.26.2	C	40.26.3	A
40.26.4	B	40.26.5	D	40.27.1	D	40.27.2	B	40.27.3	A